IMPLEMENTATION GUIDE · OCTOBER 2026

Open, hybrid AI, from the camera to the frontier

Four layers and one orchestrator: Qwen in-house, DeepSeek in-house or via API, and the frontier only when it pays off



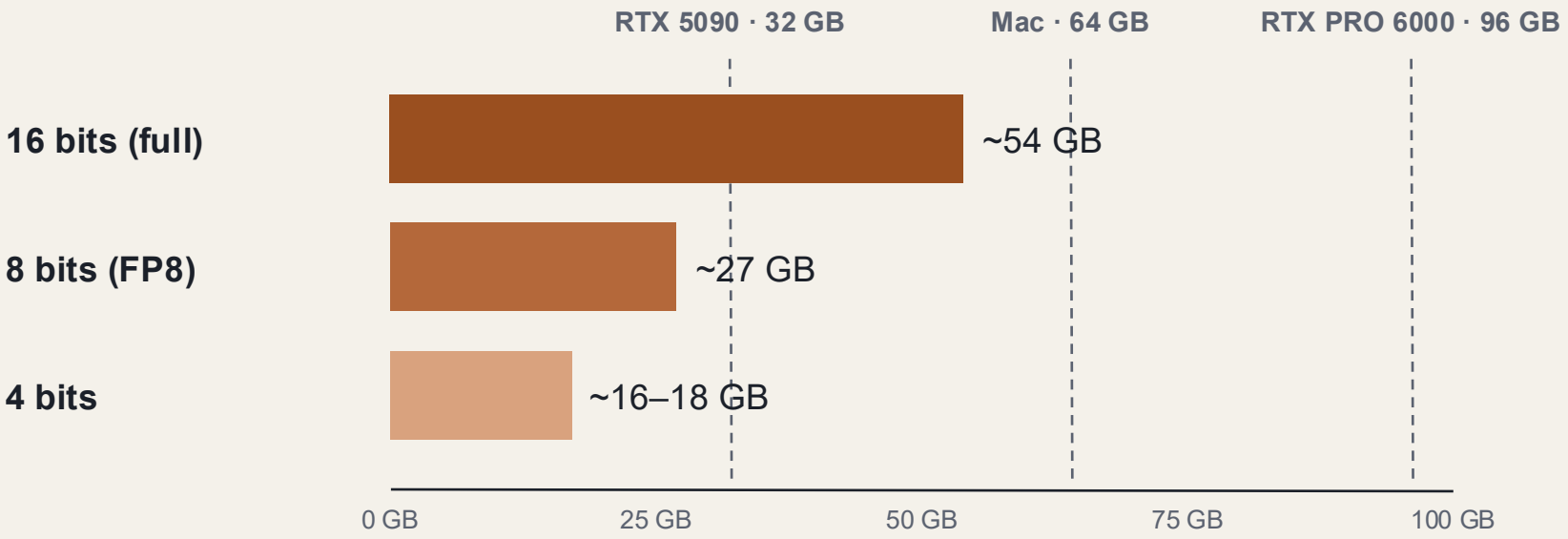
Edge Local Large open model Frontier

Marcelo José Rovai · UNIFEI · AIEng4D

Concept 1 · Does the model fit? Parameters × precision

Memory ≈ parameters × bytes per parameter + conversation cache

EXAMPLE: QWEN3.8 27B (DENSE)



Quantization

Going from 16 to 8 or 4 bits cuts memory to a half or a quarter and speeds up writing, with some loss of quality.

Not just the weights

Each conversation needs free memory for its cache (concept 4).

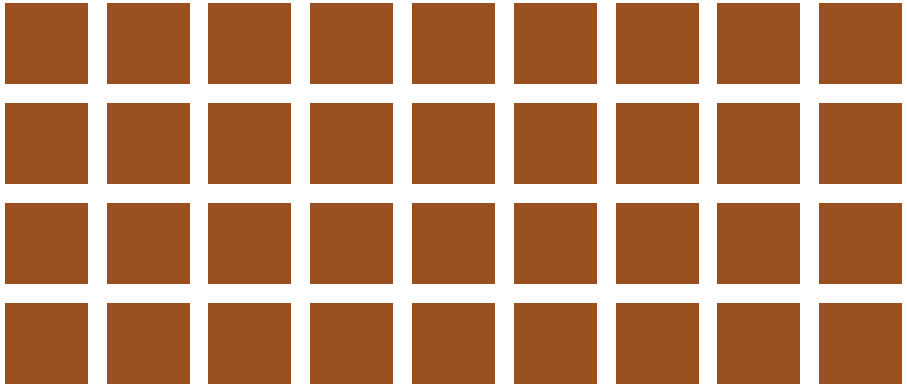
Rule of thumb: one billion parameters ≈ 2 GB at 16 bits, 1 GB at 8 bits, and ~0.6 GB at 4 bits.

Approximate sizes, weights only · Qwen3.8 27B at 8 bits: ~27 GB (video "Mac Studio vs NVIDIA RTX," Oct 2026)

Concept 2 · Dense model or mixture of experts (MoE)

DENSE

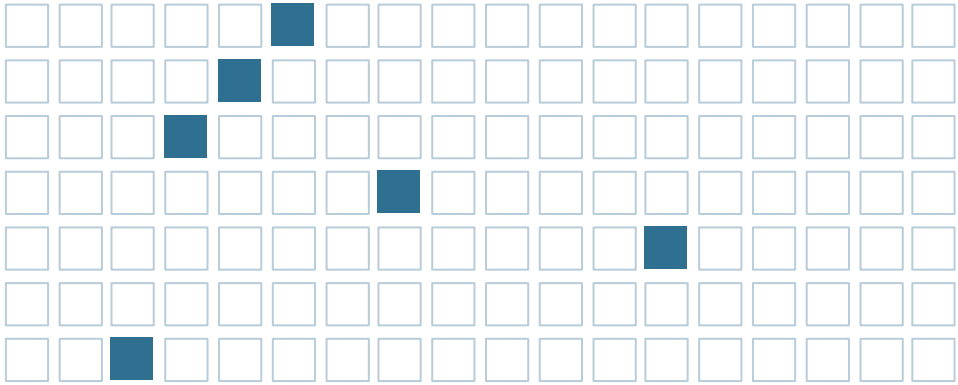
Qwen3.8 27B



27 billion parameters, all active for every token

MIXTURE OF EXPERTS (MOE)

DeepSeek V4.1 Flash



552 billion in total; each token goes through 6 of 384 experts in each layer: ~16 billion active

Memory depends on the total

Speed depends on the active ones

The next token picks other experts, so all of them must stay loaded in memory

Concept 3 · Reading and writing: prefill and decode



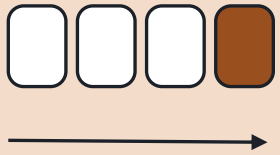
PREFILL

Read

Processes the whole context: instructions, documents, and history.

Limited by compute (TFLOPS): the GPU wins

DECODE



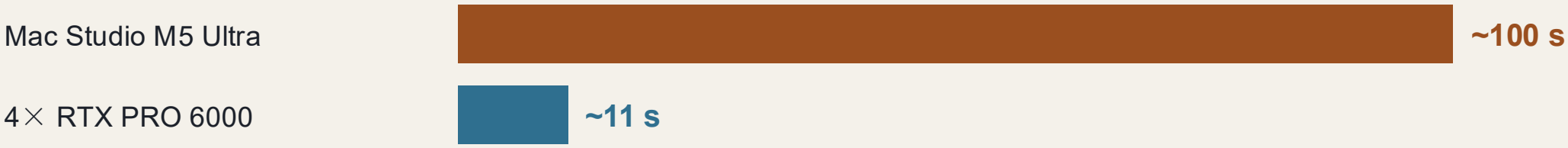
Write

Generates one token at a time; for each one it reads all the weights.

Theoretical ceiling: 1,792 GB/s ÷ 27 GB ≈ 66 tok/s

MTP (multi-token prediction) and speculative decoding can beat it: they propose several tokens and verify them at once.

Reading 100,000 tokens before the first word



An agent rereads everything on every turn: 30 turns ≈ 1 hour of reading on the Mac, ~5.5 min on the server.

Video "Mac Studio vs NVIDIA RTX," Oct 2026 · MacStories and community measurements, on similar but not identical models

Concept 4 · KV cache: each conversation's memory

ONE RTX PRO 6000 CARD · 96 GB



32 KB
per token of context (8 bits)

→ 1.86 million
tokens fit in the 59 GB

→ ≈ 14 agents
with 128,000 tokens each, at the same time

If it doesn't fit

The agent fails, or its context must be truncated, which is like erasing its memory.

Prefix cache

If the start of the conversation hasn't changed, it isn't read again: 98% in the 111-hour case.

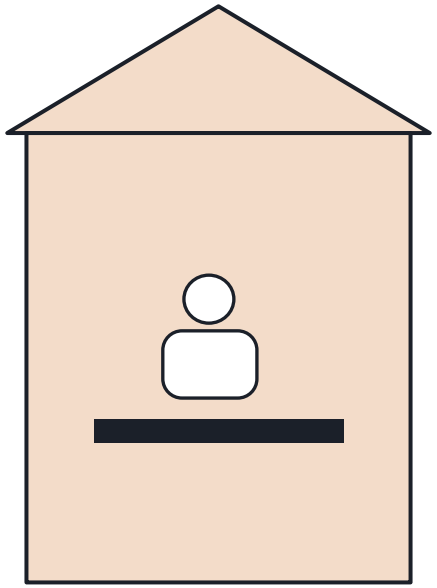
DeepSeek compresses it

One million tokens in under 1 GB, against ~800 GB for a standard transformer of the same size.

Capacity comes first: you can't serve in parallel what doesn't fit in memory.

Video "Mac Studio vs NVIDIA RTX," Oct 2026 · theoretical capacity, not performance · DeepSeek: video "What Would it Cost…," Oct 2026

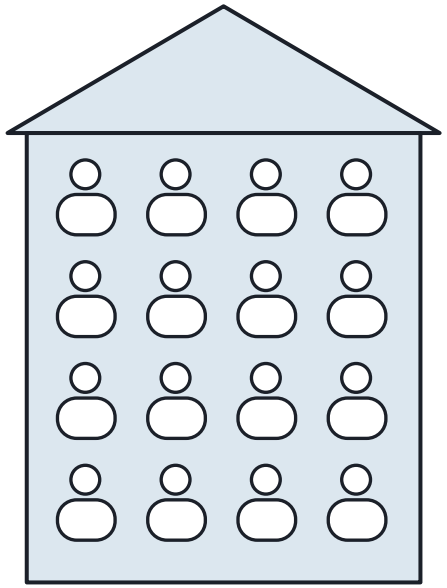
Concept 5 · One or many: the library



MAC

A huge building with a single desk

It holds a giant book (a large model), but serves one person at a time. With 3 simultaneous requests, total throughput rises only 1.23×.



GPU SERVER

A library with many desks

At each step, the weights are read once and every user gets a token: serving 32 costs almost the same as serving one (batching).

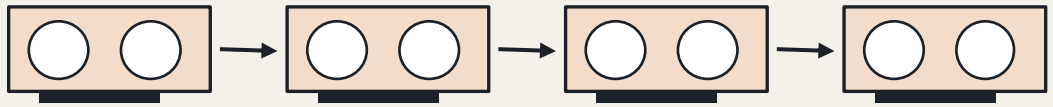
Mac: how big can the model be?

GPU: how many agents at once?

Analogy and measurements: video "Mac Studio vs NVIDIA RTX," Oct 2026 (Mac Studio M5 Ultra measured by MacStories)

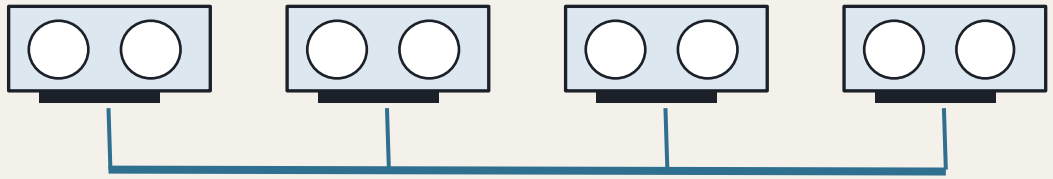
Concept 6 · Several GPUs: relay race or teamwork

BY LAYERS (PIPELINE)

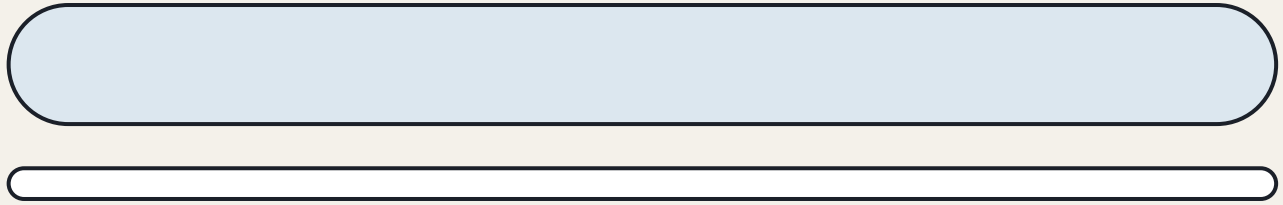


Each card waits for the previous one. 4× RTX 3090: ~70 tok/s with one, ~90–97 with two or with four.

TENSOR PARALLELISM



All cards work on every layer and sync after each one (all-reduce): the link between cards decides.



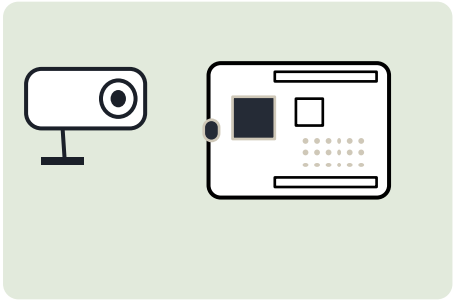
900 GB/s NVLink 4 (H100, data center)

64 GB/s per direction · PCIe 5.0 x16

Over PCIe, four cards is a sensible limit. The RTX 5090 does not support direct card-to-card communication: great alone, poor for scaling.

Video "Mac Studio vs NVIDIA RTX," Oct 2026 · NVIDIA forums (P2P on GeForce RTX 50)

Four layers, one rule: the data decides where the AI works



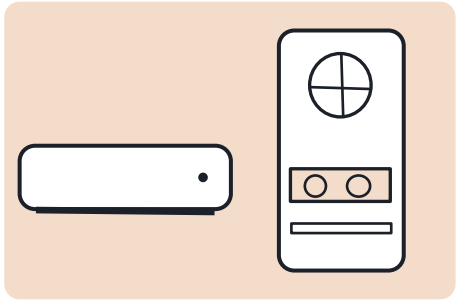
1 · EDGE

Qwen3.5 0.8B / 2B

on an Arduino UNO Q

Cameras and sensors, no internet

~US$79 per point



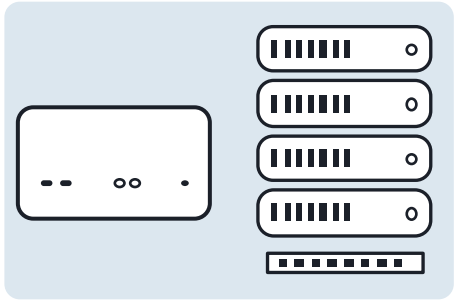
2 · LOCAL

Qwen3.8 27B

on a Mac or GPU PC

Code and sensitive data

Fixed cost: power



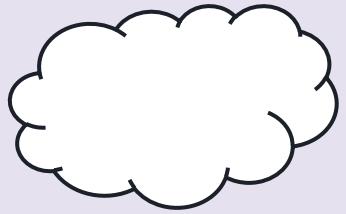
3 · LARGE OPEN

DeepSeek V4.1 Flash

in-house or via API

High volume and agents

~US$0.26 per task



4 · FRONTIER

Claude Opus 5.5

via API, by exception

Complex, occasional work

~US$6 per task

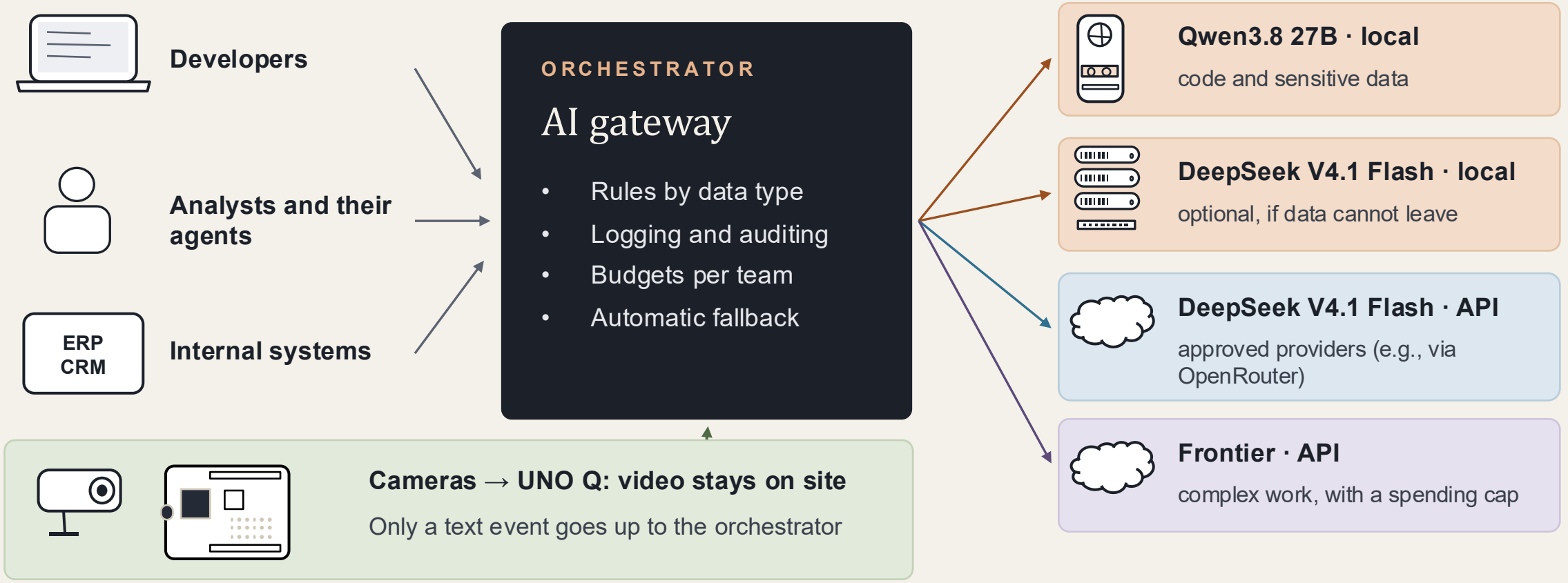
← more control and privacy

more capability and higher cost per use →

Cost per task: Artificial Analysis, Oct 2026 · UNO Q: US list price

Everything goes through an orchestrator

An AI gateway with company rules between people and models



One OpenAI-compatible interface: changing models means changing a rule, not rewriting applications.

First, classify the data

Which layer may receive each class of data

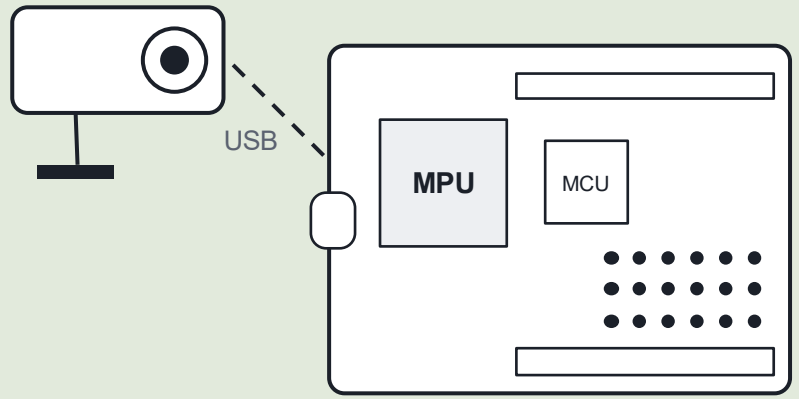
Layer	Confidential own code, customers, HR, contracts, video of people	Internal manuals, work email, catalogs	Public web, standards, published documents
Edge · UNO Q	Yes	Yes	Yes
Local · Qwen3.8 27B	Yes	Yes	Yes
DeepSeek V4.1 Flash local	Yes	Yes	Yes
DeepSeek · API (China)	No	With caution	Yes
DeepSeek · zero-retention API	Under contract	Yes	Yes
Frontier via API	No	Under contract	Yes

DeepSeek is an open model: dozens of providers serve it, in China, the US, and other countries. Choose by where each one processes data and what it does with it; under Chile's Law 21.719, every transfer of personal data must be justifiable.

Suggested classification: adapt it to your company's policy and legal advice

Layer 1 · Edge: a UNO Q watches the camera

Arduino UNO Q + USB camera + Qwen3.5 0.8B or 2B, no internet



Arduino UNO Q

Linux (Debian) + MCU STM32	2–4 GB of RAM
Model at 4 bits: ~0.5–1.2 GB	No internet · ~US$79

1. **Capture**
   The USB camera takes pictures of the gate, the warehouse, or the parking lot.
2. **Filter**
   A lightweight detector (motion, for example) discards frames where nothing changes.
3. **Describe**
   Qwen3.5 looks at the frame and puts it into words: "person next to the gate, 11:14 p.m."
4. **Alert**
   The MCU switches on a light or a relay; the board sends the event to the orchestrator.

This is not real-time video: the model looks at one frame when something changes, and takes minutes.

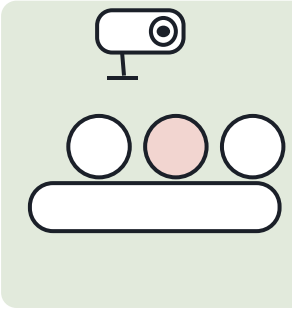
UNO Q: Qualcomm Dragonwing QRB2210 + STM32U585 · model size ≈ 0.56 GB per billion parameters at 4 bits

Layer 1 · Other edge use cases



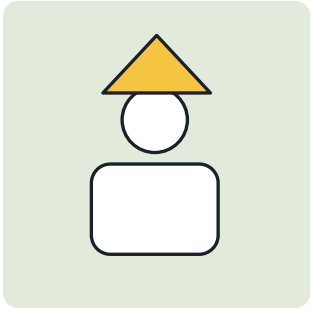
Analog gauges

A photo of a pressure gauge or water meter becomes a number, without sending the image.



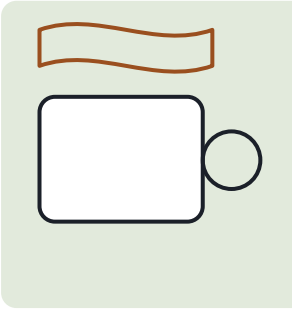
Counting and quality

Count fruit or spot defects on the packing line, right next to the belt.



Industrial safety

Hard hat and vest on? An alert before entering a hazardous area.



Maintenance

A vibration sensor and the model explain in words that something is wrong.

Edge rule: video and raw data stay on site; only text events leave.

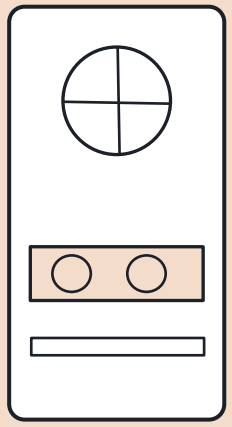
Layer 2 · Local: Qwen3.8 27B for code and sensitive data

FOR ONE PERSON



Mac with 32–64 GB

or



PC with RTX 5090 (32 GB)

34

points on the index: more than the frontier in Feb 2026 (Opus 4.6: 32)

~16–18 GB

of memory at 4 bits

Apache 2.0

free commercial use

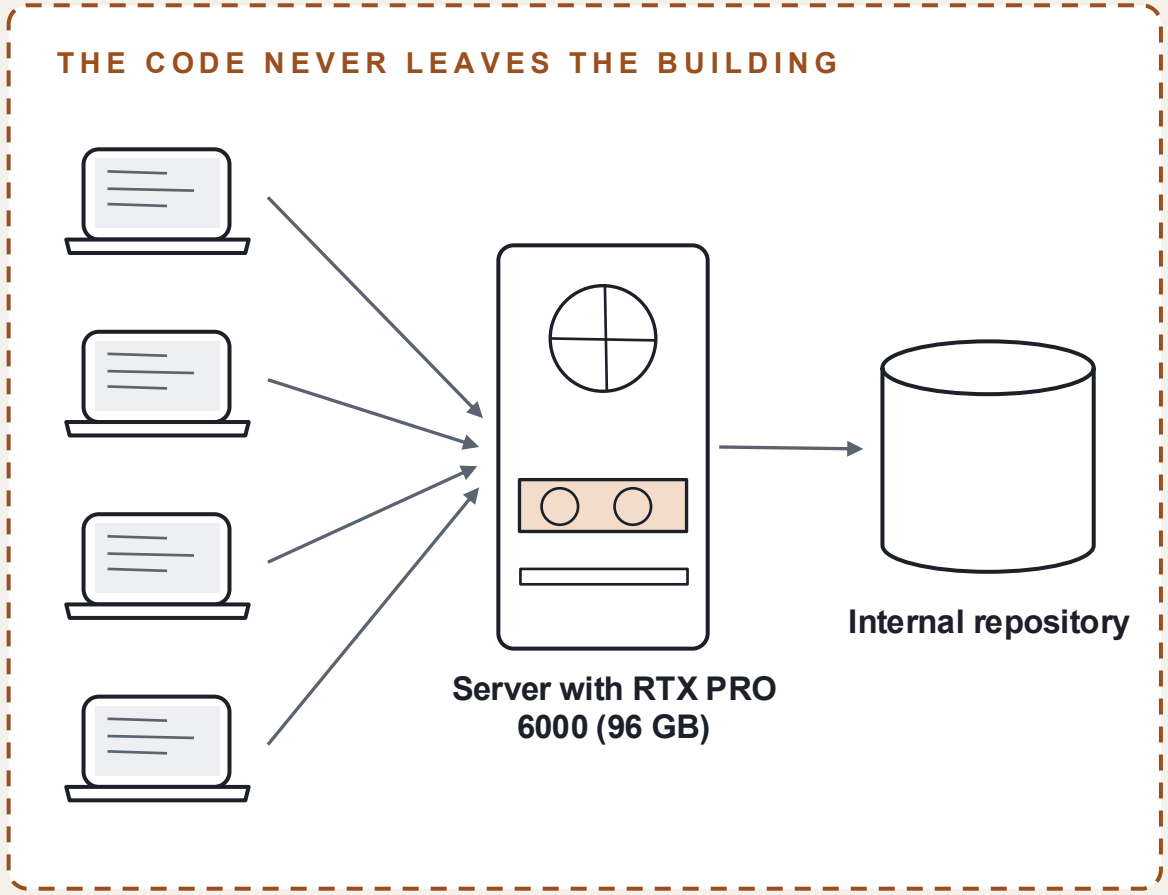
What for

- Coding assistant inside the editor
- Review of proprietary code
- Search across contracts and HR documents
- Agents with internal tools

Model	Engine	Local API	Harness
Qwen3.8 27B	llama.cpp · MLX · vLLM	OpenAI-compatible	code, documents, agents

Index: Artificial Analysis v4.3.2, Oct 2026 · memory and hardware: Annex B of the talk "Open and local AI"

Layer 2 · A server for the development team



1

Mac for one, GPU for many

One 96 GB card loads Qwen at 8 bits and leaves cache for ~14 agents with long contexts.

2

Measure with several agents

One fast agent doesn't guarantee five: measure tokens per second with 1, 4, and 8 agents at once.

3

Pin the version

In-house, the model never changes without notice; on an API, the DeepSeek V4 Flash name already points to 4.1.

A kitchen with one chef: a second order doesn't bring a second chef.

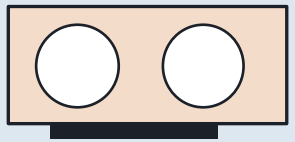
Layer 2 · Mac or GPU server?



Mac Studio M5 Ultra

96 GB · US$5,499 · 1.2 TB/s · ≤480 W

- ~3× cheaper per GB of memory
- Quiet, fits on a desk
- A regular outlet
- Holds its resale value
- Best for: one person, large models



RTX PRO 6000

96 GB · US$16,000 list, card only · 1.79 TB/s · 300 W (Max-Q)

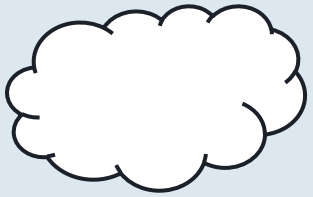
- Reads 100,000 tokens in ~11 s with 4 cards (Mac: ~100 s)
- 30 or more users at once
- KV cache for ~14 agents per card
- Tensor parallelism with P2P
- 4 Max-Q cards on a single circuit

None of this shows up in a dollars-per-gigabyte table: the Mac measures size; the GPU, concurrency.

Video "Mac Studio vs NVIDIA RTX," Oct 2026 · US list prices · reading figures on similar, not identical, models

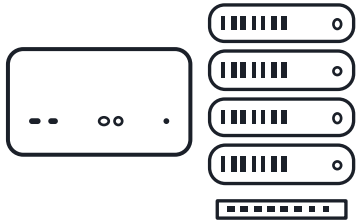
Layer 3 · DeepSeek V4.1 Flash: via API or in-house?

Index 45 · MIT license · 552 billion parameters, 16 billion active



Via API

COST	~US$0.26 per task · ~US$19 per day of heavy agent use
SPEED	~60–220 tok/s per agent, depending on the provider
DATA	Go to the provider you choose, in China or elsewhere
AVAILABILITY	DeepSeek's API: ~5 h outage on Oct 15 · with several providers: 99.9%



In-house

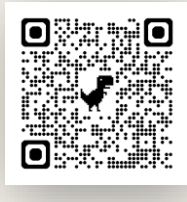
COST	Used Mac Studio 512 GB: ~US$17–23k · 4× Spark: ~US$29k
SPEED	35–90 tok/s with 1 agent · ~15 tok/s each with 16
DATA	Never leave the company
AVAILABILITY	Depends on your technical team

Start with the API for non-sensitive work; buy hardware only if data cannot leave or a team will keep it busy all day.

Artificial Analysis (cost per task) · OpenRouter (speed and availability by provider) · community measurements, Oct 2026

Layer 3 · Via API does not mean via China

DeepSeek V4.1 Flash 1017 on OpenRouter: one open model, served by 20 providers



Orchestrator →

OpenRouter

One API, many providers; if one fails, it moves to the next

Example rule: only zero-retention providers in approved countries

→

Together	**Baseten**
Fireworks	**DeepInfra**
Cloudflare	**Crusoe**
Parasail	DeepSeek · excluded
Alibaba Cloud · excluded	SiliconFlow · excluded

… and ten more

20

providers serve the same model

US$0.07–0.30

per million input tokens, by provider (output: US$0.18–1.20)

99.93%

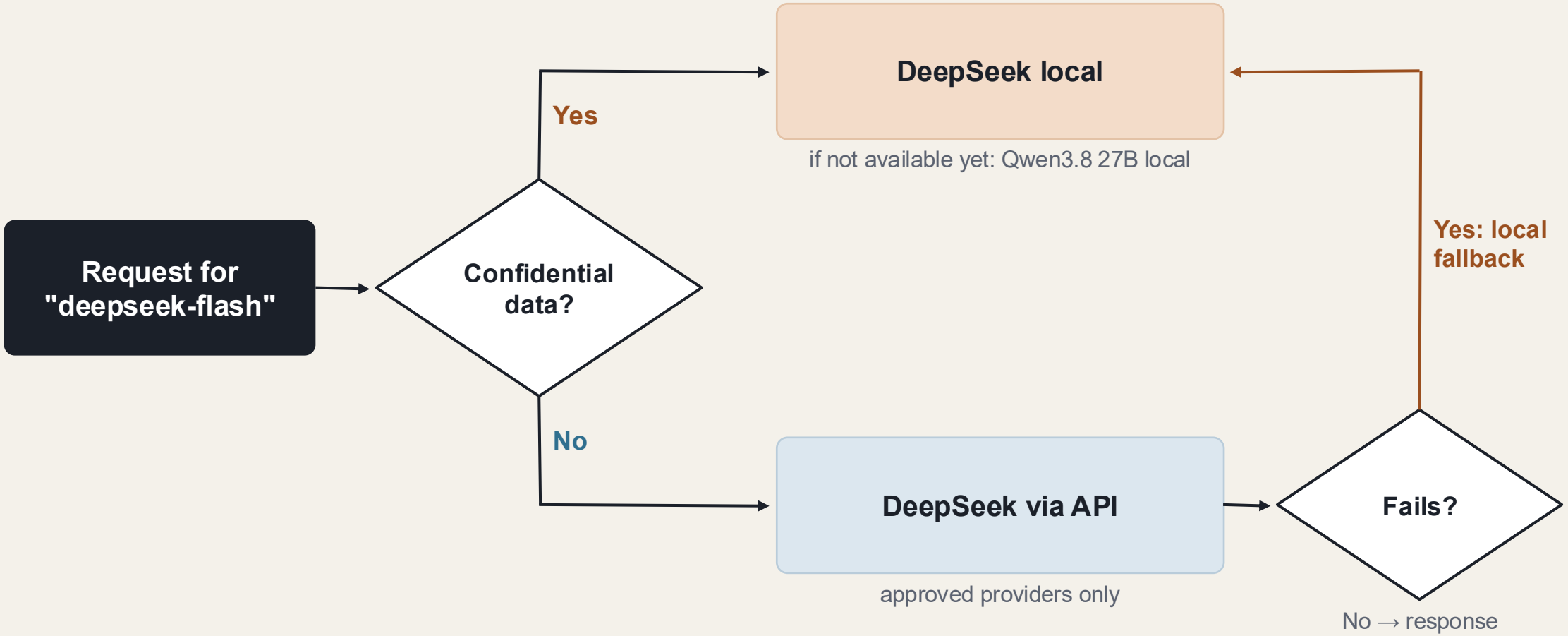
availability over 3 days with routing, against 92.45% without it

Open also means choosing who serves it, and under what rules.

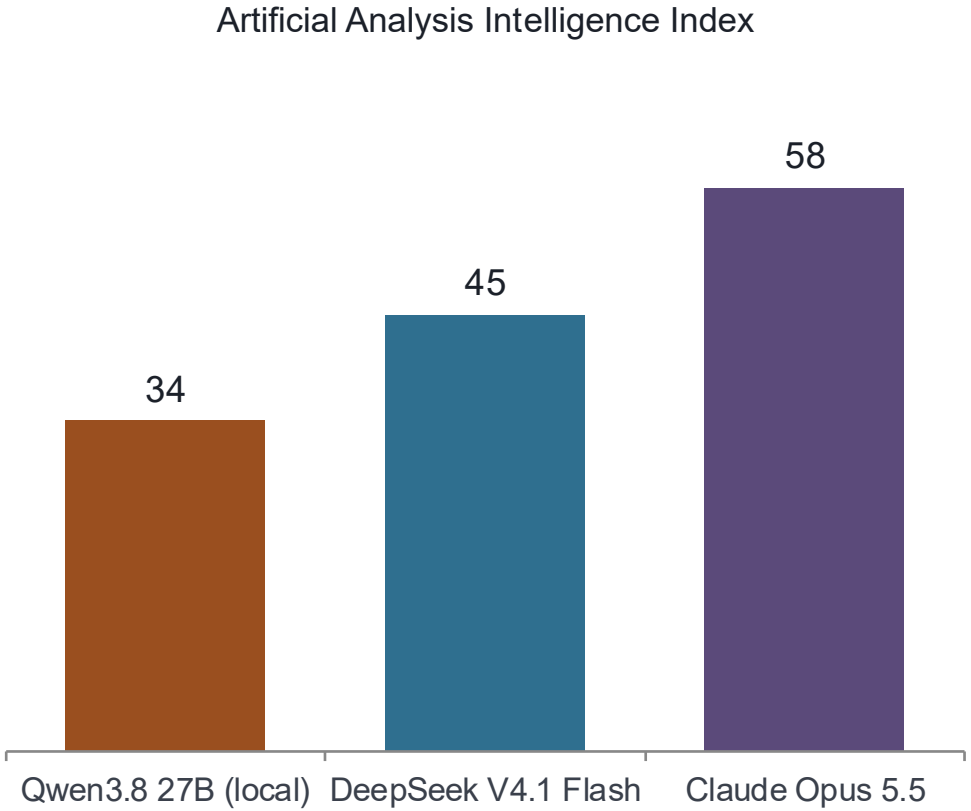
OpenRouter, model providers page, Oct 20, 2026 · the rule is an example: check each provider's data policy and country

Layer 3 · One model, two paths, one configuration

The orchestrator decides on every request; whoever sends it sees no difference



Layer 4 · Frontier: by exception and on a budget



US$5.98

per task, against US$0.26 for DeepSeek: ~23×

95 tok/s

against ~209 for DeepSeek via API

Use it for

- Strategy studies and due diligence
- System architecture and hard code
- Final review of key documents

THREE RULES

Never with confidential data · monthly spending cap · log every use

Artificial Analysis, Intelligence Index v4.3.2, Oct 2026 · cost per task and speed measured on each vendor's API

The orchestrator: simple rules, written by the company

If the request…	…it goes to
Carries camera video or images	**Edge (UNO Q): only the event leaves**
Carries proprietary code or confidential data	**Qwen3.8 27B local (or DeepSeek local)**
Is high-volume and non-sensitive: classify, summarize, extract	**DeepSeek via API, with approved providers**
Is complex, occasional, and free of sensitive data	**Frontier, with a spending cap**
Hits a provider that is down	**The next open option**

GATEWAY

LiteLLM or another open gateway

ENGINES

llama.cpp · MLX · vLLM · SGLang

REVIEW

People review the rules every month, using the usage data

An example in numbers: a 50-person company

Component	Qty	Approx. cost	What for
Arduino UNO Q + USB camera	8	~US$900	Gates, warehouses, parking
Pilot: 64 GB Mac or PC with RTX 5090	1	~US$2,700–6,000	Qwen3.8 27B for 1–3 people
Team: server with 1× RTX PRO 6000	1	~US$20,000	Qwen3.8 27B for ~14 long-context agents
Mini PC for the orchestrator	1	~US$800	Rules, logs, and dashboard
DeepSeek V4.1 Flash via API	—	~US$19 per day of heavy use	High-volume, non-sensitive
Frontier via API	—	Cap, e.g., US$500/month	Complex, occasional work
Optional: 4× DGX Spark + switch	1	~US$29,000	DeepSeek in-house, if data cannot leave

PILOT

US$4.5–8k

no large servers, plus API usage

WITH SERVER

~US$22–28k

without the DGX Spark option

Approximate US values, Oct 2026, before taxes and import · USB camera estimated at ~US$30 · RTX PRO 6000 at list price (US$16,000) plus a host computer estimated at ~US$4,000 · DeepSeek: the 111-hour coding-agent case

A 90-day implementation plan

WK 1–2

Inventory and rules

- AI uses that already exist
- Data classification
- Initial rules and gateway choice

WK 3–6

Local pilot

- Qwen3.8 27B with 3–5 developers
- Measure speed and quality
- Cache on and measured

WK 5–8

Edge pilot

- 2 UNO Qs on real cameras
- Count true and false alarms
- Tune the pre-filter

WK 7–12

API and frontier

- DeepSeek via API for high volume
- Frontier with a spending cap
- Cost dashboard; decide on large hardware

Measure from day one: cost per task · time per task · % of requests that never leave the company

Risks and controls

1 Origin and license

Qwen: Apache 2.0; DeepSeek: MIT. Weights only from the official source, with file verification.

2 Quality

Evaluate with the company's real cases; heavily compressed versions lose accuracy.

3 Operations

Someone updates, monitors, and checks that the cache works.

4 Data on APIs

Each provider processes in a different country under a different policy: the data classification decides.

5 Agent security

Least privilege and human approval for risky actions.

6 Concurrency

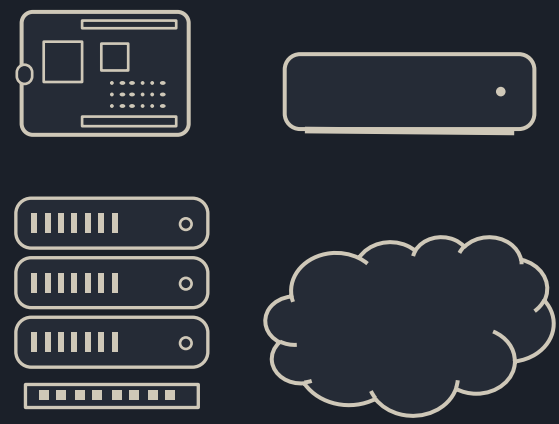
What works with one agent can choke with sixteen: measure before you buy.

Open doesn't remove risk: it trades it for risks the company can manage.

TO CLOSE

Open by default, frontier by exception

- The edge watches and alerts, with no internet
- Local protects code and sensitive data
- DeepSeek, in-house or via API, does the heavy lifting
- The frontier, only when it's worth it



If your AI provider goes dark on Monday, your company keeps running on Tuesday.

Marcelo José Rovai · mjrovai.com